DO NOW — RECALL

Symmetric and asymmetric encryption

Name: _____ Class: _____ Date: _____

1. Which technique protects data against casual interception?

2. What does a network proxy do?

3. What is the difference between confidentiality and integrity?

4. Rising-edge D flip-flop Q=0, D=1 just before next edge. What happens to Q, and when?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Encryption.

2. Model answer

Acts as an intermediary for requests/responses.

3. Model answer

Confidentiality prevents unauthorised reading; integrity detects unauthorised changes.

4. Model answer

Q changes to 1 on the next rising edge, not simply when D changes.

Vocabulary — know these terms

symmetric — Shared-secret encryption.

asymmetric — Public/private-key cryptography.

public key — Distributable key in a pair.

private key — Secret member of key pair.

TLS — Protocol establishing secure authenticated channels.

NEW LEARNING

Read, annotate and explain

Understand

Symmetric encryption uses the same secret key for encryption and decryption. The difficulty is sharing that key securely.

Trace

Asymmetric encryption uses a public/private key pair. Public-key encryption for confidentiality uses a recipient's public key to encrypt and their private key to decrypt.

Evaluate

Real secure connections typically use a hybrid protocol: authenticated asymmetric cryptography to establish secrets, then efficient symmetric encryption of application data.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

Predict first

Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

Show method

Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe symmetric and asymmetric key arrangements. [6]

Question 2

Explain how a secure website can combine asymmetric and symmetric methods. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate symmetric-only encryption for a school communicating with 200 external partners. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A public key must be kept secret to use public-key encryption.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.